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EDUCATION

ULTIMATE SOLUTION BOOK:
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Inferential Statistics for Business

KU LEUVEN • BACHELOR OF BUSINESS ADMINISTRATION

VERSION: 10 NOV 2026

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About Me

I'll make sure you pass this exam. I know exactly what it takes to get there.



BACKGROUND

- My name is **Jovan Samardzic**, I am 26 years old
- Got my **MSc in Statistics at KU Leuven**, graduated **Magna Cum Laude**
- Recipient of the **Science@Leuven Scholarship**

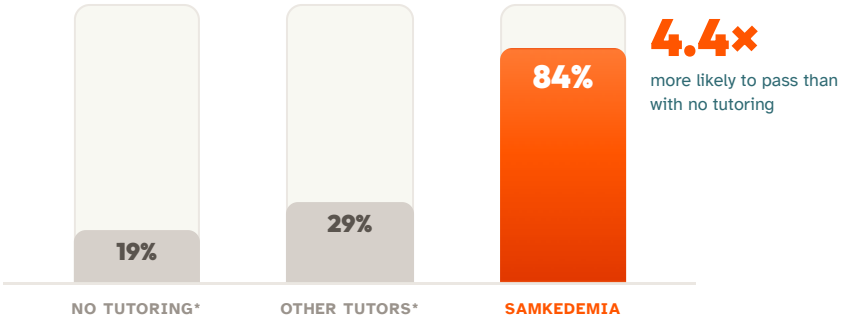
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Contents

1 Chapter 1 - Introduction **5**

Chapter 1 - Question 1 5

Chapter 1 - Question 2 10

Chapter 1 - Question 3 14

Chapter 1 - Question 4 18

Chapter 1 - Question 5 22

Chapter 1 - Question 6 24

2 Chapter 2 - Confidence Intervals **29**

3 Chapter 3 - Testing One Population Parameter **30**

4 Chapter 4 - Testing Two Population Parameters **31**

5 Chapter 5 - One-Way ANOVA **32**

6 Exam: 2016 August **33**

7 Exam: 2017 January **34**

8 Exam: 2017 August **35**

9 Exam: 2021 January 1 **36**

10 Exam: 2021 January 2 **37**

11 Exam: 2022 January **38**

12 Exam: 2023 January **39**

13 Exam: 2024 August **40**

14 Exam: 2026 January 1 **41**

15 Exam: 2026 January 2 **42**

16 Exam: 2026 August 1 **43**

17 Exam: 2026 August 2 **44**

Chapter 1 - Introduction

Question 1

A portfolio consists of a fraction p of investments in real estate and a fraction $1 - p$ of investments in stocks. The return on the investments in real estate is measured by X and the return on the investments in stocks is measured by Y. The return of the portfolio is measured by $R = pX + (1 - p)Y$.

Suppose that $\mu_X = 5\%$, $\sigma_X = 2.8\%$, $\mu_Y = 11.7\%$, $\sigma_Y = 8.6\%$, and $\rho = -0.16$.

(a) Give an economic explanation for the negative correlation.

(b) Calculate the expectation and standard deviation of R in the case $p = 0.7$.

(c) Determine the value of p at which $\text{Var}[R]$ attains its minimum. (In mathematical finance, $\text{Var}[R]$ is a measure for the risk of the portfolio.)

Solution 1: (a) Provide a brief economic explanation for the negative correlation.

Real estate and stocks compete for the same investment money. When stock returns rise, investors tend to shift money out of real estate and into stocks, which can push real estate returns down - and the other way around when stocks do poorly.

Answer: This substitution between the two asset classes is what produces the negative correlation.

Tip: Mean and Standard Deviation of a Linear Combination

For any linear combination $W = aX + bY + c$ of two random variables X and Y, the mean and standard deviation of W are:

$$\mu_W = a\mu_X + b\mu_Y + c, \qquad \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\rho_{X,Y}\,\sigma_X\sigma_Y}$$

This holds generally - whether or not X and Y are normal, correlated, or independent. The constant c never affects the spread, only the mean. If X and Y are independent, $\rho_{X,Y} = 0$ and the cross term drops out.

Solution 1: (b) Calculate the expectation and standard deviation of the portfolio return R for $p = 0.7$

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random real estate investment and a random stock investment, and record their returns.
- **Mother variables:**

X	return on a random real estate investment (in %)
Y	return on a random stock investment (in %)

- **Other random variables:**

$R = pX + (1-p)Y$	return on the combined random portfolio

- **Given:**

$\mu_X = 0.05$	population mean return of all real estate investments
$\sigma_X = 0.028$	population standard deviation of return of all real estate investments
$\mu_Y = 0.117$	population mean return of all stock investments
$\sigma_Y = 0.086$	population standard deviation of return of all stock investments
$\rho_{X,Y} = -0.16$	population correlation between all real estate and stock returns

Since the problem does not state that X and Y are normal, we can find R's mean and standard deviation directly, but cannot conclude a specific distribution for R.

Step 1: Find the mean of R.

With $a = p = 0.7$ and $b = 1 - p = 0.3$:

$$\mu_R = a\mu_X + b\mu_Y = (0.7 \cdot 0.05) + (0.3 \cdot 0.117)$$

$$= 0.035 + 0.0351 = 0.0701$$

Step 2: Find the standard deviation of R.

First we find the variance:

$$\sigma_R^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\rho_{X,Y}\,\sigma_X\sigma_Y$$

$$= (0.7)^2 \cdot (0.028)^2 + (0.3)^2 \cdot (0.086)^2 + 2 \cdot 0.7 \cdot 0.3 \cdot (-0.16) \cdot 0.028 \cdot 0.086$$

$$= 0.0008879824$$

Then take the square root:

$$\sigma_R = \sqrt{0.0008879824} \approx 0.0298$$

Answer: The expected return is $\mu_R = 0.0701$ (7.01%), and the standard deviation is $\sigma_R \approx 0.0298$ (2.98%).

Tip: Minimizing a Quadratic Function

A quadratic function $g(p) = Ap^2 + Bp + C$ with $A > 0$ opens upward, so its minimum is at its vertex:

$$p^* = -\frac{B}{2A}$$

Equivalently, this is exactly where the first derivative $g'(p) = 2Ap + B$ equals zero - both methods always agree.

Solution 1: (c) Find the fraction p that minimizes the portfolio risk (variance)

Step 1: Rewrite Var(R) as a quadratic function of p.

With $a = p$ and $b = 1 - p$:

$$\text{Var}(R) \;=\; p^2\sigma_X^2 + (1-p)^2\sigma_Y^2 + 2p(1-p)\,\rho_{X,Y}\,\sigma_X\sigma_Y$$

Substituting the given values and expanding in p:

$$\text{Var}(R) \;=\; 0.00895056\,p^2 - 0.01556256\,p + 0.007396$$

We can minimize this quadratic in two equivalent ways:

Method 1: Vertex of a parabola.

Here $A = 0.00895056$ and $B = -0.01556256$:

$$p = -\frac{B}{2A} = -\frac{-0.01556256}{2 \cdot 0.00895056} = \frac{0.01556256}{0.01790112} \approx 0.8694$$

Method 2: Using derivatives.

Take the derivative of $\text{Var}(R)$ with respect to p and set it to zero:

$$\frac{d}{dp}\text{Var}(R) = 0.01790112\,p - 0.01556256 \quad \Rightarrow \quad p = \frac{0.01556256}{0.01790112} \approx 0.8694$$

The second derivative, 0.01790112, is positive, confirming this is a minimum.

Step 3: Find the mean and standard deviation of R at this minimum-variance p.

With $a = p \approx 0.86936$ and $b = 1 - p \approx 0.13064$:

$$\mu_R \;=\; a\mu_X + b\mu_Y \;=\; (0.86936 \cdot 0.05) + (0.13064 \cdot 0.117) \;\approx\; 0.05875$$

$$\sigma_R^2 \;=\; 0.00895056\,p^2 - 0.01556256\,p + 0.007396 \;\approx\; 0.0006311$$

$$\sigma_R \;\approx\; \sqrt{0.0006311} \;\approx\; 0.02512$$

Answer: The portfolio risk is minimized at $p \approx 0.8694$, i.e. about 86.94% invested in real estate. At this p, $\mu_R \approx 0.05875$ (5.875%) and $\sigma_R \approx 0.02512$ (2.512%).

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Question 2

Let X measure the taxable income of the wife in a randomly selected heterosexual married couple and Y measure the taxable income of the husband in the same married couple. Assume that (X, Y) has a bivariate normal distribution with

$$\mu_X = €1734, \quad \sigma_X = €366, \quad \mu_Y = €1921, \quad \sigma_Y = €453, \quad \rho_{X,Y} = 0.45.$$

(a) What is the probability distribution of the total taxable income of a randomly chosen married couple?

(b) Women must pay a tax of 30% and men a tax of 40%. What is the probability that the net income of the wife is larger than the net income of the husband in a randomly selected married couple?

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c, and $W = aX + bY + c$:

$$\left.\begin{array}{l} X \sim \mathcal{N}(\mu_X, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array}\right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\sigma_X\sigma_Y\,\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

Solution 2: (a) What is the probability distribution of the total taxable income of a randomly chosen married couple?

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random heterosexual married couple and record the taxable incomes of the wife and husband.
- **Mother variables:**

X	taxable income of the wife in a random married couple (in €)
Y	taxable income of the husband in a random married couple (in €)

- **Other random variables:**

$T = X + Y$	total taxable income of a random married couple

- **Given:** (X, Y) is bivariate normal, with population parameters:

$\mu_X = 1734$	population mean taxable income of all wives
$\sigma_X = 366$	population standard deviation of taxable income of all wives
$\mu_Y = 1921$	population mean taxable income of all husbands
$\sigma_Y = 453$	population standard deviation of taxable income of all husbands
$\rho_{X,Y} = 0.45$	population correlation between all wives' and husbands' incomes

Since T is a linear combination with $a = 1$, $b = 1$, $c = 0$, it is also exactly normal - we just need its mean and standard deviation:

$$\mu_T = 1734 + 1921 = 3655$$

$$\sigma_T = \sqrt{366^2 + 453^2 + 2 \cdot 366 \cdot 453 \cdot 0.45} \approx 698.84$$

Answer: The total taxable income follows $T \sim \mathcal{N}(3655,\ 698.84)$ (in €).

Tip: Comparing Two Variables

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

Solution 2: (b) What is the probability that the net income of the wife is larger than the net income of the husband?

Step 1: Rewrite the comparison as a statement about the difference.

The wife pays 30% tax, so her net income is $0.7X$; the husband pays 40% tax, so his net income is $0.6Y$. We move everything to one side:

$$P(0.7X > 0.6Y) = P(0.7X - 0.6Y > 0)$$

Define $D = 0.7X - 0.6Y$, the difference in net incomes.

Step 2: Find the distribution of the difference.

This is a linear combination with $a = 0.7$, $b = -0.6$, $c = 0$:

$$\mu_D = 0.7 \cdot 1734 - 0.6 \cdot 1921 = 61.2$$

$$\sigma_D = \sqrt{(0.7)^2 \cdot 366^2 + (-0.6)^2 \cdot 453^2 + 2 \cdot 0.7 \cdot (-0.6) \cdot 366 \cdot 453 \cdot 0.45} \approx 277.20$$

So the difference is also normally distributed:

$$D \sim \mathcal{N}(61.2,\ 277.20)$$

Step 3: Compute the probability.

Since μ_D is positive, the wife's net income is higher than the husband's on average, so we expect

$P(D > 0)$ above 0.5:

$$P(D > 0) = \texttt{normalcdf}(0, \texttt{E99}, 61.2, 277.20) \approx 0.5874$$

Answer: The probability that the wife's net income is larger than the husband's is 0.5874.

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Question 3

The weights of butter dishes that leave some factory are normally distributed with mean $250\,\text{g}$ and standard deviation $4\,\text{g}$. Let X and Y measure the weights of two independently and randomly chosen butter dishes from the production.

(a) What is the value of $P(X > Y)$?

(b) Calculate $P(|X - Y| > 4)$.

(c) Determine $P(\max(X, Y) < 250)$.

Tip: Comparing Two Variables

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c, and $W = aX + bY + c$:

$$\left.\begin{array}{l} X \sim \mathcal{N}(\mu_X,\, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y,\, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array}\right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W,\, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\sigma_X\sigma_Y\,\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

Solution 3: (a) What is the value of $P(X > Y)$?

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random butter dish from the factory and record its weight.
- **Mother variables:**

X	weight of a random butter dish (in g)
Y	weight of a second, independent random butter dish (in g)

- **Other random variables:**

$D = X - Y$	difference in weight between the two random butter dishes (in g)

- **Given:**

$\mu_X = \mu_Y = 250$	population mean weight of all butter dishes
$\sigma_X = \sigma_Y = 4$	population standard deviation of weight of all butter dishes

X, Y are independent

Step 1: Translate the comparison into a statement about the difference.

We move everything to one side:

$$P(X > Y) = P(X - Y > 0)$$

Define $D = X - Y$. Since X and Y are independent, this is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho_{X,Y} = 0$.

Step 2: Find the distribution of D, and compute the probability.

Plug in the given means and standard deviations:

$$\mu_D = 250 - 250 = 0$$

$$\sigma_D = \sqrt{4^2 + 4^2} = \sqrt{32} \approx 5.657$$

So $D \sim \mathcal{N}(0,\ 5.657)$. Since this distribution is symmetric around 0, the probability of a value above 0 is exactly half:

$$P(X > Y) = P(D > 0) = \texttt{normalcdf}(0, \texttt{E99}, 0, 5.657) = 0.5$$

Answer: $P(X > Y) = 0.5$.

Solution 3: (b) Calculate $P(|X - Y| > 4)$

Step 1: Translate the event into two one-sided pieces.

"Differs by more than 4 g" doesn't say which dish is heavier - either X could exceed Y by more than 4 g, or Y could exceed X by more than 4 g. These are two separate, non-overlapping events, so we add their probabilities instead of using absolute value notation:

$$P(D > 4) + P(D < -4)$$

where $D = X - Y \sim \mathcal{N}(0,\ 5.657)$ from part (a).

Step 2: Compute the probability.

By symmetry, both tails carry the same probability:

$$P(D > 4) = \texttt{normalcdf}(4, \texttt{E99}, 0, 5.657) \approx 0.23975$$

$$P(D < -4) = \texttt{normalcdf}(-\texttt{E99}, -4, 0, 5.657) \approx 0.23975$$

Adding them:

$$P(D > 4) + P(D < -4) \approx 0.23975 + 0.23975 = 0.4795$$

Answer: $P(|X - Y| > 4) \approx 0.4795$.

Tip: Probability of a Maximum

The statement "the maximum of two values is less than a number" is a concise way of saying that **both** of the individual values must be less than that number.

$$P(\max(X, Y) < a) = P(X < a \text{ and } Y < a)$$

Furthermore, if X and Y are independent, this simplifies to the product of their individual probabilities:

$$P(\max(X, Y) < a) = P(X < a) \cdot P(Y < a)$$

Solution 3: (c) Determine $P(\max(X, Y) < 250)$

Step 1: Rewrite the maximum condition as a product.

Since the choices of butter dishes are independent, we can multiply their probabilities:

$$P(\max(X, Y) < 250) = P(X < 250 \text{ and } Y < 250) = P(X < 250) \cdot P(Y < 250)$$

Step 2: Compute the probability.

First, we find the probability for a single dish, $P(X < 250)$. Since the distribution $X \sim N(250,\ 4)$ is symmetric around its mean of 250, the probability of being less than the mean is exactly 0.5:

$$P(X < 250) = 0.5$$

Now we can calculate the final probability:

$$P(\max(X, Y) < 250) = 0.5 \cdot 0.5 = 0.25$$

Answer: The probability that the maximum weight of the two dishes is less than 250 g is 0.25.

Question 4

The random pair (W, L) measures the (width, length) of a random rectangular piece of building site and is bivariate normally distributed with $\mu_W = 30\,\text{m}$, $\sigma_W = 2\,\text{m}$, $\mu_L = 50\,\text{m}$, $\sigma_L = 3\,\text{m}$, and correlation $\rho_{W,L}$. The calculated perimeter is $X = 2(W + L)$.

(a) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = 0$.

(b) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = +0.5$.

(c) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = -0.5$.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c, and $W = aX + bY + c$:

$$\left.\begin{array}{l} X \sim \mathcal{N}(\mu_X,\, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y,\, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array}\right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W,\, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\sigma_X\sigma_Y\,\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

Solution 4: (a) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = 0$

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random rectangular building site and record its width and length.
- **Mother variables:**

W	width of a random building site (in m)
L	length of the same random building site (in m)

- **Other random variables:**

$X = 2W + 2L$	perimeter of the random building site (in m)

- **Given:** (W, L) is bivariate normal, with population parameters:

$\mu_W = 30$	population mean width of all building sites
$\sigma_W = 2$	population standard deviation of width of all building sites
$\mu_L = 50$	population mean length of all building sites
$\sigma_L = 3$	population standard deviation of length of all building sites
$\rho_{W,L}$	population correlation between width and length of all building sites

Bivariate normal just means W and L are each individually normal, linked by $\rho_{W,L}$. Because of this, any linear combination of them - including the perimeter $X = 2W + 2L$ - is itself exactly normal too, no matter how strongly W and L are correlated. This is what makes all three parts solvable with `normalcdf`, once we find X's mean and standard deviation.

Step 1: Identify a, b, and c.
The perimeter is $X = 2W + 2L$, so $a = 2$, $b = 2$, $c = 0$.

Step 2: Find the mean.
The mean never depends on the correlation, so this value is the same for every part of this exercise. First the general formula, then plug in:

$$\mu_X = a\mu_W + b\mu_L + c = 2\mu_W + 2\mu_L = 2 \cdot 30 + 2 \cdot 50 = 160$$

Step 3: Find the standard deviation, as a general function of $\rho_{W,L}$.
Unlike the mean, the standard deviation does depend on the correlation - so instead of plugging

in a number right away, we keep $\rho_{W,L}$ symbolic, to reuse this same formula in parts (b) and (c):

$$\begin{aligned}\sigma_X(\rho_{W,L}) &= \sqrt{a^2\sigma_W^2 + b^2\sigma_L^2 + 2ab\,\sigma_W\sigma_L\,\rho_{W,L}} \\ &= \sqrt{2^2\cdot 2^2 + 2^2\cdot 3^2 + 2\cdot 2\cdot 2\cdot 2\cdot 3\,\rho_{W,L}} \\ &= \sqrt{16 + 36 + 48\,\rho_{W,L}} = \sqrt{52 + 48\,\rho_{W,L}}\end{aligned}$$

So:

$$X \;\sim\; \mathcal{N}\big(160,\ \sqrt{52 + 48\,\rho_{W,L}}\big)$$

Step 4: Plug in $\rho_{W,L} = 0$ and compute the probability.

Substitute $\rho_{W,L} = 0$ into the formula from Step 3:

$$\sigma_X(0) = \sqrt{52 + 48\cdot 0} = \sqrt{52} \quad\Rightarrow\quad X \;\sim\; \mathcal{N}(160,\ \sqrt{52})$$

Now we find the probability:

$$P(X \le 165) = \texttt{normalcdf}(-\texttt{E99}, \texttt{165}, \texttt{160}, \sqrt{52}) \approx 0.7560$$

Answer: $P(X \le 165) \approx 0.7560$.

Solution 4: (b) Calculate $P(X \le 165)$ in case $\rho_{W,L} = 0.5$

Step 1: Plug in $\rho_{W,L} = 0.5$.

We reuse $X \;\sim\; \mathcal{N}\big(160,\ \sqrt{52 + 48\,\rho_{W,L}}\big)$ from part (a):

$$\sigma_X(0.5) = \sqrt{52 + 48\cdot 0.5} = \sqrt{76} \quad\Rightarrow\quad X \;\sim\; \mathcal{N}(160,\ \sqrt{76})$$

Step 2: Compute the probability.

With X's distribution in hand:

$$P(X \leq 165) = \texttt{normalcdf}(-\texttt{E99}, \texttt{165}, \texttt{160}, \sqrt{76}) \approx 0.7169$$

Answer: $P(X \leq 165) \approx 0.7169$.

Solution 4: (c) Calculate $P(X \leq 165)$ in case $\rho_{W,L} = -0.5$

Step 1: Plug in $\rho_{W,L} = -0.5$.

We reuse $X \sim \mathcal{N}\left(160, \sqrt{52 + 48\,\rho_{W,L}}\right)$ from part (a):

$$\sigma_X(-0.5) = \sqrt{52 + 48 \cdot (-0.5)} = \sqrt{28} \quad \Rightarrow \quad X \sim \mathcal{N}(160, \sqrt{28})$$

Step 2: Compute the probability.

With X's distribution in hand:

$$P(X \leq 165) = \texttt{normalcdf}(-\texttt{E99}, \texttt{165}, \texttt{160}, \sqrt{28}) \approx 0.8276$$

Answer: $P(X \leq 165) \approx 0.8276$.

Question 5

In an elevator, there is a warning: "maximum 500 kg or 6 persons." Suppose that the body weights are normally distributed with mean 75 kg and standard deviation 15 kg. The body weight of a random person can be considered as independent of the body weights of other randomly chosen people.

What is the probability that 6 randomly chosen people weigh more than 500 kg?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left.\begin{array}{l} X_1, \ldots, X_n \sim \mathcal{N}(\mu,\, \sigma) \\ \text{independent} \end{array}\right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu,\, \sqrt{n}\,\sigma) \\ \overline{X}_n \sim \mathcal{N}\left(\mu,\, \dfrac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Solution 5:

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random person and record their body weight.
- **Mother variable:**

  X weight of a random person (in kg)

- **Other random variables:**

  $W_6 = X_1 + \cdots + X_6$ total weight of 6 random people (in kg)

- **Given:**

$\mu = 75$	population mean weight of all people
$\sigma = 15$	population standard deviation of weight of all people
$n = 6$	sample size

$X \sim \mathcal{N}(\mu = 75,\ \sigma = 15)$

$X_1, \ldots, X_6$ are independent

Step 1: Find the distribution of the total weight.

The weight of one person is normal. We're looking for the sum of 6 people's weights, so:

$$\left.\begin{array}{l} X_1, \ldots, X_6 \sim \mathcal{N}(75,\ 15) \\ \\ \text{independent} \end{array}\right\} \quad \Rightarrow \quad W_6 \sim \mathcal{N}(6 \cdot 75,\ \sqrt{6} \cdot 15) = \mathcal{N}(450,\ \sqrt{1350})$$

Step 2: Compute the probability.

Now that we know the distribution of W_6, we can find the probability that it exceeds 500 kg:

$$P(W_6 > 500) = \texttt{normalcdf}(500, \texttt{E99}, 450, \sqrt{1350}) \approx 0.0868$$

Answer: The probability that 6 randomly chosen people weigh more than 500 kg is 0.0868.

Question 6

In a barbershop, the durations of the haircuts of men are normally distributed with mean 21 minutes and standard deviation 6 minutes, and the durations of the haircuts of women are normally distributed with mean 48 minutes and standard deviation 12 minutes. The durations of different haircuts can be assumed to be independent.

(a) There are 3 male customers in this barbershop. What is the probability that the total duration of their haircuts exceeds 1.5 hours?

(b) What is the probability that the average duration of the haircuts of 3 men is larger than 30 minutes?

(c) What is the probability that the total duration of the haircuts of 5 men is smaller than the total duration of the haircuts of 2 women?

Tip: Sum and Average of a Normal Variable

If we add up or average n independent normal variables, the result is also normal:

$$\left.\begin{array}{l} X_1, \ldots, X_n \sim \mathcal{N}(\mu,\, \sigma) \\ \text{independent} \end{array}\right\} \Rightarrow \begin{array}{l} W_n \sim \mathcal{N}(n\mu,\, \sqrt{n}\,\sigma) \\ \overline{X}_n \sim \mathcal{N}\left(\mu,\, \dfrac{\sigma}{\sqrt{n}}\right) \end{array}$$

This holds exactly, for any n - no minimum sample size needed, since the X_i are already normal.

Solution 6: (a) Probability the total duration of 3 men's haircuts exceeds 1.5 hours

Before computing anything, we set up the framework:

- **Random experiment:** Pick a random male customer and a random female customer, and record their haircut durations.

- **Mother variables:**

X	duration of a random man's haircut (in minutes)
Y	duration of a random woman's haircut (in minutes)

- **Other random variables:**

W_3	total duration of 3 random men's haircuts (in minutes)
$\overline{X}_3$	mean duration of 3 random men's haircuts (in minutes)
W_{5M}	total duration of 5 random men's haircuts (in minutes)
W_{2W}	total duration of 2 random women's haircuts (in minutes)
$D = W_{5M} - W_{2W}$	difference between the men's total and the women's total (in minutes)

- **Given:**

$\mu_X = 21$	population mean duration of all men's haircuts
$\sigma_X = 6$	population standard deviation of duration of all men's haircuts
$\mu_Y = 48$	population mean duration of all women's haircuts
$\sigma_Y = 12$	population standard deviation of duration of all women's haircuts

$X \sim \mathcal{N}(\mu_X = 21,\ \sigma_X = 6)$
$Y \sim \mathcal{N}(\mu_Y = 48,\ \sigma_Y = 12)$
all haircut durations are independent

Note that 1.5 hours is 90 minutes. The duration of one man's haircut is normal. We're looking for the sum of 3 men's haircuts, so:

$$\left.\begin{array}{l} X_1, X_2, X_3 \sim \mathcal{N}(21,\ 6) \\ \text{independent} \end{array}\right\} \Rightarrow \quad W_3 \sim \mathcal{N}(3 \cdot 21,\ \sqrt{3} \cdot 6) = \mathcal{N}(63,\ 10.3923)$$

Now we find the probability that it exceeds 90 minutes:

$$P(W_3 > 90) = \texttt{normalcdf}(90, \texttt{E99}, 63, 10.3923) \approx 0.0047$$

Answer: The probability that the total haircut time of 3 men exceeds 1.5 hours is 0.0047.

Solution 6: (b) Probability the mean duration of 3 men's haircuts is larger than 30 minutes

This time we're looking for the mean instead of the sum, using the same 3 men's haircuts:

$$\left.\begin{array}{l} X_1, \ldots, X_3 \sim \mathcal{N}(21,\ 6) \\ \text{independent} \end{array}\right\} \quad \Rightarrow \quad \overline{X}_3 \sim \mathcal{N}\left(21,\ \frac{6}{\sqrt{3}}\right) = \mathcal{N}(21,\ 3.4641)$$

Now we find the probability that it exceeds 30 minutes:

$$P(\overline{X}_3 > 30) = \texttt{normalcdf}(30, \texttt{E99}, 21, 3.4641) \approx 0.0047$$

This matches part (a) exactly - "total exceeds 90 minutes" and "mean exceeds 30 minutes" describe the exact same event, since $90/3 = 30$.

Answer: The probability that the mean haircut time for 3 men exceeds 30 minutes is 0.0047.

Tip: Comparing Two Variables

To find a probability comparing two variables (e.g. $P(X > Y)$), move everything to one side first, turning it into a statement about their difference:

$$P(X > Y) = P(X - Y > 0)$$

Then find the distribution of $X - Y$ using the linear combination formula for two normal variables.

Tip: Linear Combination of Two Normal Variables

For any constants a, b, c, and $W = aX + bY + c$:

$$\left.\begin{array}{l} X \sim \mathcal{N}(\mu_X,\, \sigma_X) \\ Y \sim \mathcal{N}(\mu_Y,\, \sigma_Y) \\ \rho_{X,Y} = \text{correlation} \end{array}\right\} \Rightarrow \begin{array}{l} W \sim \mathcal{N}(\mu_W,\, \sigma_W) \\ \mu_W = a\mu_X + b\mu_Y + c \\ \sigma_W = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\,\sigma_X\sigma_Y\,\rho_{X,Y}} \end{array}$$

In other words: a linear combination of two normal variables is itself normal.

Solution 6: (c) Probability the total duration of 5 men's haircuts is smaller than that of 2 women's

Step 1: Find the distribution of the total duration of 5 men's haircuts.

The duration of one man's haircut is normal. We're looking for the sum of 5 men's haircuts, so:

$$\left.\begin{array}{l} X_1, \ldots, X_5 \sim \mathcal{N}(21,\, 6) \\ \text{independent} \end{array}\right\} \Rightarrow W_{5M} \sim \mathcal{N}(5 \cdot 21,\, \sqrt{5} \cdot 6) = \mathcal{N}(105,\, 13.4164)$$

Step 2: Find the distribution of the total duration of 2 women's haircuts.

The duration of one woman's haircut is normal too. We're looking for the sum of 2 women's haircuts, so:

$$\left.\begin{array}{l} Y_1, Y_2 \sim \mathcal{N}(48,\, 12) \\ \text{independent} \end{array}\right\} \Rightarrow W_{2W} \sim \mathcal{N}(2 \cdot 48,\, \sqrt{2} \cdot 12) = \mathcal{N}(96,\, 16.9706)$$

Step 3: Find the distribution of the difference, and compute the probability.

We want $P(W_{5M} < W_{2W})$, so we rewrite it as a statement about the difference:

$$P(W_{5M} < W_{2W}) = P(W_{5M} - W_{2W} < 0)$$

Define $D = W_{5M} - W_{2W}$. This is a linear combination with $a = 1$, $b = -1$, $c = 0$, and $\rho = 0$ since the two sums are independent:

$$\mu_D = 105 - 96 = 9$$

$$\sigma_D = \sqrt{13.4164^2 + 16.9706^2} \approx 21.6333$$

So $D \sim \mathcal{N}(9,\ 21.6333)$. Now we find the probability:

$$P(D < 0) = \texttt{normalcdf}(-\texttt{E99}, 0, 9, 21.6333) \approx 0.3387$$

Answer: The probability that the total haircut time of 5 men is smaller than that of 2 women is 0.3387.

Chapter 2 - Confidence Intervals

Chapter 3 - Testing One Population Parameter

Chapter 4 - Testing Two Population Parameters

Chapter 5 - One-Way ANOVA

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Exam: 2017 January

Exam: 2017 August

Exam: 2021 January 1

Exam: 2021 January 2

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